

BRAC UNIVERSITY
Department of Computer Science and Engineering
CSE423: Computer Graphics

Assignment 02 | Total Marks: 20 | Summer 2025

Submission: July 25, 2025

- Q1.**
- a.** What is the significance of the AND operation in the Cohen-Sutherland Algorithm? [2]
 - b.** Find the coordinates (x, y) of the point $t = 0.3$ for the line (1, 3) to (7, 10) using the parametric line equation. [2]
 - c.** A viewing window from (-50, -10) to (60, 100) is given. Check whether the line segment (-50, -70) to (40, 100) is trivially accepted or trivially rejected or partially inside. If partially inside then **find the clipping coordinates** using:
 - A. Cyrus-Beck Algorithm. [4]
 - B. Cohen-Sutherland Algorithm [4]
- Q2.**
- a.** Explain how reflection about an arbitrary line differs from reflection about the coordinate axes. [2]
 - b.** In a scene from Loki's timeline, the Time Variance Authority agents need to adjust the coordinates of the Nexus Point, initially located at (70, 50), to coincide perfectly with a temporal anomaly at (100, 150). To achieve this, they first scale the point uniformly by a factor of 2 about (75, 25), then rotate it 45° counter-clockwise around (50, 50), and finally shear it along both x and y directions using unknown shear factors sh_x and sh_y with respect to the point (10, 10). Determine the values of sh_x and sh_y that ensure the point arrives at (100, 150). Use homogeneous coordinates, show the matrix composition in the given order, and solve step by step. [6]